GPMeeting614

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1 Goals

1.1 How and when does MOGP improve or worsen calibration?

We plan to analyse calibration differences between MOGP and single-band GP along four axes. Together these factors explain not just *whether* MOGP helps, but *why* and *under what conditions*.

- **Object class.** Different transient classes exhibit different light-curve morphologies, spectral energy distributions, and inter-band correlation structures. We therefore ask not only which classes benefit from MOGP, but why. Candidate explanations include the smoothness of the light curve (which affects how well a stationary kernel can model it), the typical degree of inter-band correlation (which controls how much information auxiliary bands carry about the target band), and average data sparsity per class. To ground the analysis, we compare class-level calibration gains (ΔNLPD, $\Delta|z_{\text{std}} - 1|$) against measurable class-level statistics such as the mean learned wavelength covariance between bands and per-class sparsity profiles.

- **Band and cross-band information transfer.** In the current product-kernel model $k((t,\lambda),(t',\lambda')) = k_{\text{time}}(t,t') \cdot k_{\text{wavelength}}(\lambda,\lambda')$, cross-band transfer is governed implicitly by the wavelength kernel: bands whose central wavelengths are close to the target band receive higher covariance and should therefore contribute more to posterior variance reduction. We make this explicit through two complementary analyses. First, we extract the learned wavelength covariance matrix evaluated at the central wavelengths of all bands and inspect whether its structure is physically consistent (e.g. whether spectrally adjacent bands are more correlated). Second, we perform a *band-wise ablation*: we predict the target band r using progressively richer auxiliary-band sets (r only; $r+g$; $r+g+i$; …; all bands) and record the cross-band posterior variance reduction

$$R_{\text{cross}}^{(j\to r)}(t) = \frac{\sigma^2(y_r^* \mid D_r) - \sigma^2(y_r^* \mid D_r, D_j)}{\sigma^2(y_r^* \mid D_r) + \varepsilon}, \tag{1}$$

where D_r denotes target-band training data and D_j auxiliary-band data from band j. $\sigma^2(y_r^*|D_r)$ is the predictive variance from the single-band GP, conditioned only on D_r. $\sigma^2(y_r^*|D_r, D_{aux})$ is the predictive variance from **MOGP**. ϵ is a small numerical guard against division by zero. $R_{cross}(t_i) \in [0, 1]$ when MOGP reduces variance (the normal case).

We test whether $R_{\text{cross}}^{(j\to r)}$ correlates with $k_{\text{wavelength}}(\lambda_j, \lambda_r)$: if the wavelength kernel is well specified, spectrally closer bands should reduce uncertainty more. If this relationship is weak or non-monotone, it suggests kernel misspecification or that inter-band correlations are not primarily driven by spectral proximity. To isolate the calibration gain from the data-quantity effect, the **independent-band control** should be added as a third comparison point alongside single-band GP and MOGP. R_{cross} also can be redefined as $\sigma^2_{independent_control} - \sigma^2_{MOGP}$.

For each object (m), we aggregate $R_{\text{cross}}(t)$ across all held-out test points to obtain a single object-level statistic. This quantity represents the average fractional reduction in predictive variance achieved by MOGP relative to the single-band GP (or, alternatively, the independent-band control) over the object's test set. As such, it provides a summary measure of the extent to which cross-band information reduces predictive uncertainty for a given object.

We will compute the calibration gain ΔNLPD_m or any other calibration metrics, such as CRPS.

$$\Delta\text{NLPD}_m = \text{NLPD}_{single} - \text{NLPD}_{MOGP}$$

Where $\text{NLPD}_m = -\frac{1}{n_m^{test}} \sum_i \log p(y_{m,i}|\mu_{m,i}, \sigma_{m,i})$. Positive NLPD_m means MOGP is better.

Afterwards, we can create a scatter plot in which each point represents one object (m). The x-axis shows the object-level aggregated R_{cross}, while the y-axis shows the corresponding calibration gain ΔNLPD. This plot addresses the question of whether the uncertainty reduction achieved by MOGP is actually associated with improved calibration.

To summarize this relationship quantitatively, we can compute a correlation coefficient, such as Spearman's rank correlation or Pearson correlation. This analysis should be performed separately within each n_{aux} stratum. The key comparison is whether the correlation remains meaningful across all strata. If a clear relationship only appears in the dense-auxiliary-data stratum, then the observed effect is more likely driven by data quantity rather than by physically meaningful cross-band transfer learned through the wavelength kernel.

In a later stage, a Linear Model of Coregionalization (LMC) or an empirical inter-band correlation analysis could provide a more explicit parameterisation of cross-band transfer and allow a direct comparison with the calibration gains observed here.

- **Data sparsity.** Fixing object class and target band, we investigate how calibration differences between MOGP and single-band GP vary with the number of available target-band training points n_m^{target}. Within each class, objects are split into tertiles by n_m^{target} (sparse, medium, dense), and calibration metrics are compared across tertiles. The core hypothesis is that MOGP provides the largest calibration gain in the sparse regime, where the target band alone carries insufficient information to constrain the posterior, but auxiliary bands can compensate through cross-band transfer. Conversely, in the dense regime, the single-band GP already has sufficient data, and the marginal benefit of auxiliary bands is expected to diminish. Stratifying within class controls for the confound that different classes have different typical sparsity levels, isolating sparsity as the variable of interest.

- **Failure cases and negative transfer.** Cross-band conditioning does not guarantee calibration improvement. If the wavelength kernel misestimates inter-band correlations, or if auxiliary bands carry conflicting or weakly correlated signals, MOGP may reduce predictive variance inappropriately, producing overconfident posteriors with degraded coverage and $z_{\text{std}} > 1$. We explicitly test for this by identifying objects and classes where $\Delta\text{NLPD} < 0$ or coverage worsens under MOGP, and examining whether such cases are concentrated in particular sparsity regimes, classes, or band combinations. (This turns potential failure modes into interpretable findings rather than anomalies to be suppressed.)

1.2 How to improve the calibration of MOGP

It depends on the analysis from the first goal. We've already know that forcing peak into training data can enhance the performance. Some other promising methods include:

- Data Augmentation.
- Post-hoc calibration: PIT histogram shows whether the hyperparameters overfit, then we could use temperature-scaling to vary the standard deviation of GP.
- Data Sampling (not necessary): Latin hypercube sampling
- Anomaly Detection.

2 Calibration Methods

2.1 Probability Integral Transform (PIT) Histogram

A calibration diagnostic for probabilistic forecasts. Given a cumulative distribution function (CDF) F and observed value y, the PIT value is $z = F(y)$.

According to probability integral transform, If the forecast is perfectly calibrated, the collection of PIT values should be uniformly distributed. Consider carrying out the stratified PIT analysis, where the objects are splitted into bins by training set size(e.g., sparse/medium/dense tertiles), class, and band. If the model is calibrated, each bin should contain approximately the same fraction of points.

Problem: Investigate how they handle this issue. (Binning)

2.2 Scalar Calibration Scores

Besides visual histograms, which is hard to aggregate across strata and thus make the result harder to analyze, the scalar calibration scores could be a cleaner calibration approach. Moreover, PIT-based diagnostics such as KS-PIT and IAE focus solely on calibration, while calibration scores like CRPS provide a single global measure of predictive performance[2]. Below lists some scalar calibration scores.

Continuous Ranked Probability Score (CRPS): The CRPS loss is usually written as

$$\mathrm{CRPS}(F, x) = \int_{-\infty}^{\infty} \left(F(y) - \mathbf{1}_{\{y \geq x\}}\right)^2 dy$$

It compares the predicted CDF and the observed CDF at all possible thresholds, then sum up the squared difference. It evaluates the whole predictive distribution. We have RMSE checking the mean prediction, coverage checking whether x falls inside selected intervals, and PIT checking calibration, but does not reward sharpness directly. One of the advantages of CRPS is that it checks both, whether the distribution is centered near the observation, the uncertainty width, and whether the full CDF is close to the prediction. CRPS is more robust and easier to interpret compared to NLPF/LogS, which can be unstable when σ_i is very small. [1]

2.3 Normalized RMSE

In the current evaluation metrics, raw RMSE is hard to compare across objects because PLAsTiCC objects have very different flux scales. Therefore, a normalized RMSE would help in making RMSE comparable across objects. I adopt the formula below:

$$\mathrm{NRMSE}_m = \frac{\sqrt{\frac{1}{n_m}\sum_i (y_{m,i}^{test} - \hat{y}_{m,i})^2}}{s_m}$$

where

$$s_m = max\left(P_{95}\left(|y_{m,all}|\right), \epsilon\right)$$

Note that the scale is only used for post-hoc evaluation, not for training, hyperparameter optimization, model fitting, or prediction. Hence it uses an all-observed-data object-level empirical flux scale.

2.4 Calibration Among Classes

To investigate the model performance across different classes, aggregate the object-level evaluation metrics by object class.

From the experiment on 500 objects, the single-band GP shows clear class-dependent behavior.

Table 1: Class-wise accuracy and sharpness of the single-band GP.

Class	N_{obj}	$\bar{n}_{\text{train}}$	Median n_{train}	RMSE	NRMSE	NLPD
AGN	27	23.93	19	39.59	0.33	4.14
EB	24	31.42	41	419.79	9.72	7.71
KN	9	17.78	17	3.95	0.82	2.91
M-dwarf	39	31.54	41	11.64	3.12	6.79
RRL	5	36.00	41	2521.22	2.26	10.43
SLSN-I	15	25.87	20	5.97	0.69	3.16
SNII	95	24.72	20	7.08	1.61	3.27
SNIa	182	27.37	20	11.75	2.80	3.50
SNIa-91bg	13	20.46	19	4.63	0.87	3.19
SNIax	13	30.92	41	4.27	1.43	2.59
SNIbc	35	24.83	19	9.12	1.54	3.20
TDE	42	17.52	17	13.51	1.79	3.73

1. EB and RRL are problematic classes. the mean RMSE is very large, about 420, and the normalized RMSE is also extremely large, about 9.72. That means this is not just a flux-scale issue. The GP is genuinely doing badly relative to the object scale. Calibration is also poor: $z_std_mean \approx 1.45$, 2σ coverage is only 0.88, and 3σ coverage is 0.925, both below the Gaussian ideal values of about 0.954 and 0.997. So for EB, the model is both inaccurate and overconfident. RRL only has 5 objects so we need more samples before making a firm class-level claim.

2. For the supernova-like classes, the GP is more stable. They mostly have much lower RMSE/NLPD/CRPS than EB/RRL. But their calibration is often in the opposite direction: many have $z_std_mean < 1$. It implies the model is underconfident.

3. The situation is mixed for AGN and M-dwarf. They have one of the better z-score standard deviations, suggesting that the average uncertainty scale is not badly wrong. However, their empirical coverage is below the nominal Gaussian levels, so the predictive distribution is still not fully calibrated and appears mildly overconfident or heavy-tailed. We need PIT histogram for further examination.

The results suggest that class morphology, temporal variability, and flux-scale heterogeneity strongly affect GP calibration and accuracy. However, conclusions for rare classes such as RRL, KN, SNIax, and SNIa-91bg should be treated cautiously because of the small number of objects.

3 Comparison between the single- and multi-band GP

For both single-GP and MOGP models, the peak per object is preserved. Though the maximum training point across all available bands is the preserved peak for MOGP. This is meaningful for time centering, which is only for visualization since non-stationary kernels are invariant to this change. Also, the quantile-based local scale is then computed given the preserved peak value. Overall, the peak preservation makes the models easier to learn the trend. Missing peak values could lead to failure in prediction.

3.1 Model A: Single-band GP

It has been proven before that the use of global and local scale, as well as the background reduction do not have a recognizable effect on single-band GP. So we opt to use local scale and background reduction. The curves can be scaled back to its original scales for downstream tasks like classification.

In the following MOGP models, we temporarily **disable the background estimator** for a clearer interpretation.

Here are some distinction of training set and scale:

- A/B: same target-only training set, same target scale.
- C/D/E: same multiband training set, same target scale
- F: fixed total budget, same target scale

3.2 Model B: Target-only MOGP

Use the multi-band GP implementation, but train only on the target-band training points with fixed wavelength equal to the target band. I expect it to approximately match the single-band GP if preprocessing and kernel settings are consistent.

3.3 Model C: auxiliary-data-per-band MOGP

Train on the same target-band training points plus selected non-target-band training points. It aims at evaluating how much each non-target band contributes to the calibration and predictive performance of a target-band prediction. Specifically, each object's observations are splitted into train/test first. While single-band GP uses target-band training poitns only, Model C uses target-band training points plus sampled auxiliary-band training points. The auxiliary-data ratio is defined relative to the target-band training set size. *Quote: Because different bands have different cadences and different missingness patterns. If one band naturally has 50 points and another has 8, then "use 50% of each band" gives very different absolute amounts of auxiliary data. The*

comparison becomes partly about band sampling density rather than controlled auxiliary information.

$$n_{aux,band} = r \cdot n_{target\,train}$$

If $n_{aux\,train} \leq n_{target\,train}$, $n_{aux,band} = n_{aux\,train}$.

3.4 Model D: Independent-band-control MOGP

Train on exactly the same points as C, but disable cross-band information transfer by using a band-independent kernel or block-diagonal covariance across bands. Other-band points should not affect target-band predictions. This experiment aims at recognizing the effect of cross-band covariances on the results.

3.5 Model E: Shuffled-wavelength-control MOGP

Train on exactly the same points as C, but randomly shuffle the wavelength labels among non-target-band observations, keeping the target-band labels fixed. Repeat several times with different seeds if possible. This experiment is to test whether the wavelength information matters for the results.

3.6 Model F: fixed total budget

With the same number of data points, one from a single band, and the other from the remaining 5 bands. For the multi-band GP, the global scale is the maximum quantile-based value from all bands.

3.7 Ablation analysis of single-band and multi-band GP models

The ablation is evaluated on the r band using 100 object-band cases and 502 held-out observations. Since different objects can contribute different numbers of held-out points, the object-weighted metrics are used as the primary comparison. Observation-weighted metrics show broadly similar trends, but they are more sensitive to objects with many test points.

Model A and Model B give numerically identical results. This is an important sanity check: the target-only MOGP implementation reduces to the single-band GP when only the target-band training points are used. Therefore, differences between the later MOGP models and the single-band baseline are not caused by an inconsistency between the single-band and multi-output GP code paths.

Among the multi-band models, Model C, the real-wavelength MOGP, gives the best overall performance. The test incorporates 50% of training points from g band and 100% of i band. Compared with the single-band baseline, it reduces the object-weighted RMSE from 11.98 to 9.95, the NLPD from 3.58 to 3.41, and the CRPS from 10.09 to 7.60. This corresponds to an RMSE

reduction of about 16.9% and a CRPS reduction of about 24.7%. This suggests that adding auxiliary-band observations with their true wavelength labels can improve target-band prediction.

However, the control models show that the improvement should be interpreted carefully. Model D, the independent-band-control MOGP, also improves RMSE relative to the single-band baseline. This does not necessarily indicate physical cross-band transfer, because Model D uses the same enlarged multiband training set as Model C, whereas Models A and B use only target-band training points. Therefore, the difference between Model D and Models A/B partly reflects the effect of data quantity and the multiband training setup. The more relevant comparison for isolating cross-band covariance is between Model C and Model D, since they use the same training points but differ in whether cross-band information transfer is enabled.

Model E, the shuffled-wavelength-control MOGP, performs worse than Model C in NLPD and CRPS, even though its RMSE is still better than the single-band baseline. Since many bands share similar light-curve features, having more data points from other bands can also benefit the model performance. This is useful evidence that the real wavelength labels matter, especially for probabilistic scoring. The real-wavelength model is not merely benefiting from the presence of additional data points; it seems to benefit from meaningful band information.

Model F, the fixed-total-budget model, performs clearly worse than all other models. Its RMSE, NRMSE, NLPD, and CRPS are all substantially larger than those of the single-band baseline. This confirms that spreading a fixed training budget across multiple bands can harm target-band prediction when the target band becomes undersampled. Therefore, auxiliary-band information does not automatically compensate for losing target-band observations.

From the calibration perspective, none of the models is perfectly calibrated. Models A and B have object-weighted $\mathrm{Std}(z) \approx 0.718$, and Model C has $\mathrm{Std}(z) \approx 0.683$, both below the ideal value of 1. This suggests that the predictive uncertainties are somewhat too wide on average, i.e. the models are underconfident at the object-averaged level. The empirical one-sigma coverages are also above the nominal Gaussian value of approximately 0.683. Model F has $\mathrm{Std}(z)$ closer to 1, but its two- and three-sigma coverages are worse, and its accuracy is much poorer, so it should not be considered better calibrated overall.

Overall, the experiment supports three conclusions. First, the single-band and target-only MOGP implementations are consistent. Second, auxiliary-band observations can improve target-band prediction when real wavelength information is used, with Model C giving the strongest overall performance. Third, the current evidence for true cross-band transfer is promising but not fully isolated, because the independent-band control also differs from the single-band baseline.

Relative changes compared with Model A. Model C reduces object-weighted RMSE by approximately 16.9%, NLPD by 0.17, and CRPS by approximately 24.7% compared with the single-band GP. Model D also reduces RMSE

Table 2: Object-weighted performance and calibration metrics for the GP ablation study on the r band. Lower RMSE, NRMSE, NLPD, and CRPS are better. For ideal Gaussian calibration, $\mathrm{Cov}_{1\sigma} \approx 0.683$, $\mathrm{Cov}_{2\sigma} \approx 0.954$, $\mathrm{Cov}_{3\sigma} \approx 0.997$, and $\mathrm{Std}(z) \approx 1$.

Model	Description	RMSE ↓	NRMSE ↓	NLPD ↓	CRPS ↓	$\mathrm{Cov}_{1\sigma}$	$\mathrm{Cov}_{2\sigma}$	$\mathrm{Cov}_{3\sigma}$	$\mathrm{Std}(z)$
A	Single-band GP	11.98	1.56	3.58	10.09	0.770	0.948	0.977	0.718
B	Target-only MOGP	11.98	1.56	3.58	10.09	0.770	0.948	0.977	0.718
C	Real-wavelength MOGP	**9.95**	**1.30**	**3.41**	**7.60**	0.786	0.968	0.990	0.683
D	Independent-band control	10.28	1.33	3.63	10.64	0.793	0.962	0.981	0.666
E	Shuffled-wavelength control	10.60	1.41	3.62	11.89	0.805	0.969	0.987	0.649
F	Fixed total budget	17.77	2.54	4.84	15.36	0.744	0.913	0.950	0.919

by approximately 14.1%, but its NLPD and CRPS are worse than Model A, suggesting that the improvement is not consistently probabilistic. Model E reduces RMSE by approximately 11.5%, but its CRPS becomes worse than Model A, indicating that shuffled wavelength labels damage distributional quality. Model F increases RMSE by approximately 48.4% and CRPS by approximately 52.2%, showing that a fixed total budget spread across bands is harmful for target-band prediction.

4 Existing Problems

1. The wavelength length scale often goes to the upper bound 2000, or the lower bound 20, implying that the model may not be learning a stable physical relationship. Further check on the wavelength kernel matrix is important. LMC could be a better solution because it avoid using the wavelength length scale, but learn a band-to-band covariance matrix. However, it has the risk of overfitting with sparse lightcurve, so it's more reasonable to learn/share parameters across many objects or objects in the same class.

2. I am struggling to disentangle the effect of cross-band transfer from the effect of simply having more training data. – How to compare single-band and multi-band GP fairly?

3. Per-class analysis on MOGP too complex. Consider the features, bands, and data availability of objects from different classes. The test only tells us the general performance of single-GP on different classes. It's good for data sparsity test where we fix object class, but what else?

4. Which analyses are most scientifically meaningful?

5 Possible Future Work

Post-hoc Calibration.
Data Augmentation.
Data Sampling (Latin hypercube sampling).
Anomaly Detection.
Sensitivity Analysis

Band-to-band correlation
LMC

6 Meeting Summary

Future work should focus more on:

- Understand under which scenario the model is overconfident or underconfident, and be able to explain the reasons(boundary values, features, nosify data...).
- Carry out post-hoc calibration.

Some interesting experiments or ideas to be considered:

- Aggregate the evaluation metrics of supernova-like classes.
- Look into spectral mixture kernel.
- Visualization idea: the number of training points as x axis, and the evaluation metrics like CRPS, NLPD as y axis. Compare Model C and D's performance curves.

Comprehend the scalar calibration scores.

References

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[2] Aurélien Pion and Emmanuel Vazquez. Design-marginal calibration of gaussian process predictive distributions: Bayesian and conformal approaches, 2025.